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Resting-state EEG alpha/theta power ratio discriminates early-onset Alzheimer's disease from healthy controls

Yağmur Özbek [a,1], Ezgi Fide [a,1], Görsev G. Yener [b,c,*]

[a] Department of Neurosciences, Institute of Health Sciences, Dokuz Eylul University, Izmir, Turkey
[b] Izmir Biomedicine and Genome Center, Dokuz Eylul University Health Campus, Izmir, Turkey
[c] Izmir University of Economics, Faculty of Medicine, Izmir, Turkey

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HIGHLIGHTS

- Resting-state EEG can be used as a valuable tool to evaluate brain dynamics in AD and healthy elderly.
- Alpha/theta ratio is a promising marker for distinguishing early-onset Alzheimer's disease (EOAD) from healthy controls.
- EOAD patients display more widespread and severe electrophysiological abnormalities.

ABSTRACT

Objectives: The present study aims to compare early-onset Alzheimer's disease (EOAD) patients with healthy controls (HC), and late-onset Alzheimer's disease (LOAD) patients using resting-state delta, theta, alpha, and beta oscillations and provide a cut-off score of alpha/theta ratio to discriminate individuals with EOAD and young HC.
Methods: Forty-seven individuals with EOAD, 51 individuals with LOAD, and demographically-matched 49 young and 51 older controls were included in the study. Spectral-power analysis using Fast-Fourier Transformation (FFT) is performed on resting-state electroencephalography (EEG) data. Delta, theta, alpha, and beta oscillations compared between groups and Receiver Operating Characteristic (ROC) curve analysis was conducted.
Results: Compared to healthy controls individuals with EOAD showed an increase in slow frequency bands and a decrease in fast frequency bands. Frontal alpha/theta power ratio is the best discriminating value between EOAD and young HC with the sensitivity and specificity greater than 80% with area under the curve (AUC) 0.881.
Conclusions: EOAD display more widespread and severe electrophysiological abnormalities than LOAD and HC which may reflect more pronounced pathological burden and cholinergic deficits in EOAD. Additionally, the alpha/theta ratio can discriminate EOAD and young HC successfully.
Significance: This study is the first to report that resting-state EEG power can be a promising marker for diagnostic accuracy between EOAD and healthy controls.
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1. Introduction

Alzheimer's Disease (AD) is the most common cause of dementia, characterized by cognitive impairment interfering with daily functioning. Based on the cut-off age of 65 years clinically categorized as early-onset AD (EOAD) and late-onset AD (LOAD). Although EOAD is rare compared to late-onset dementia, it is the most common cause of early-onset dementia and accounts for about 30–40% of these cases (Ferreira et al., 2018). In 30% of EOAD cases, non-amnestic symptoms are presented first (Rossor et al., 2010). On the contrary, LOAD's first symptom is episodic memory impairment. As the disease progresses, memory impairment is followed by deteriorations in other cognitive domains (Kaiser et al., 2014; Koedam et al., 2010).

A limited number of studies investigated neuropsychological profiles of patients with EOAD and LOAD. Several studies have

* Corresponding author at: Izmir University of Economics, Faculty of Medicine, Izmir, Turkey.
E-mail address: gorsev.yener@ieu.edu.tr (G.G. Yener).
[1] These authors contributed equally to the work.

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shown that, while memory impairments are common in LOAD, impairments in attention, language, and visuospatial abilities are more prominent in EOAD (Galton et al., 2000; Greicius et al., 2002; Koedam et al., 2010; Smits et al., 2012). However, in some studies, it is reported that the age of onset does not relate to impaired cognitive areas (Seltzer and Sherwin, 1983; Grady et al., 1987). Magnetic Resonance Imaging (MRI) studies showed that neocortical atrophy in EOAD is greater than LOAD (Ishii et al., 2005; Möller et al., 2013; Hamelin et al., 2015; Migliaccio et al., 2015). Nevertheless, hippocampal atrophy is reported to be more dominant in LOAD (Frisoni et al., 2007; Möller et al., 2013; Cavedo et al., 2014; Hamelin et al., 2015). In addition, Kaiser et al. (2012) showed EOAD patients had greater hypometabolism in posterior brain regions compared with LOAD. Yet, some studies indicating EOAD and LOAD had a similar atrophy pattern and the difference in brain volumes likely due to age-related atrophy (Eckerström et al., 2018). The main pathological features that distinguish AD from other dementias are the accumulation of senile plaques and neurofibrillary tangles in the brain (Bigio et al., 2002). Although this pathology occurs in all cases, it has been shown that its neuropathological burden is higher in patients with EOAD (Marshall et al., 2007; Eckerström et al., 2018) and the course of the disease is faster (Grønning et al., 2012; Eckerström et al., 2018) compared to patients with LOAD.

The different presentation of the disease in EOAD sometimes may cause greater delays in diagnosis (Mendez, 2017). A recent study showed that EOAD had prolonged the process between the symptom onset and diagnosis in comparison with LOAD (van Vliet et al., 2013). EOAD presents with more marked neuropsychiatric symptoms than cognitive symptoms. In early-onset dementia, initial non-dementia diagnosis is quite common and often misdiagnosed as psychiatric disorders, i.e., major depression and/or personality disorders (Mendez, 2006; van Vliet et al., 2013; Micanovic and Pal, 2014). Delays in diagnosis or misdiagnosis may have negative effects on individuals, families and caregivers such as divorce, job loss, and failure to receive proper medical treatment. The timely diagnosis gives families and individuals an opportunity to adjust to disease and its outcomes (van Vliet et al., 2011, 2013).

Electroencephalography (EEG) is a low-cost, non-invasive and useful technique in reflecting functional changes in the cerebral cortex. EEG biomarkers, have an advantage of better temporal resolution compared to other imaging techniques in investigating neurodegeneration (Cassani et al., 2018). Different EEG techniques are being used to obtain information about the disease pathology. Entropy measures the diversity of the phase relationship of two EEG signals (Lee et al., 2017; Shin et al., 2020; Alù et al., 2020). Previously, using multiscale entropy analysis, Mizuno and colleagues (Mizuno et al., 2010) showed that the EEG signal complexity of individuals with EOAD decreased at faster scales and increased at slower scales, which were related to higher and lower relative power, respectively. In the LOAD group, lower spectral entropy was found over posterior areas compared with healthy controls (Abásolo et al., 2006). Another technique that examines the instantaneous topographic maps of EEG is microstate analysis (Michel and Koenig, 2018). Conducting microstate analysis, Tait and colleagues (Tait et al., 2020) reported that individuals with LOAD had altered fronto-parietal working memory/attention networks compared to healthy elderly controls. Individuals with AD also displayed decreased synchronization likelihood in high frequencies compared with healthy controls (Nardone et al., 2018). Moreover, higher global field delta and theta power as well as lower power of alpha and beta were found in the AD group in comparison with the controls (Huang et al., 2000).

Many researchers addressed EEG power analysis for AD to find specific alterations (Moretti et al., 2004; Jeong, 2004; Osipova et al., 2005; Babiloni et al., 2006a; Dauwels et al., 2010; Hsiao et al., 2014; Vecchio et al., 2013; Koelewijn et al., 2017; Cassani et al., 2018; Rossini et al., 2020). Resting-state EEG studies indicated that the background alpha rhythm is slowed and posterior alpha amplitudes decreased with increased age (Klass and Brenner, 1995; Klimesch, 1999; Babiloni et al., 2006a). In line with alpha slowing, a global increase in delta and theta power is also reported (Ishii et al., 2017). The slowing of alpha, amplitude reduction of alpha and beta, and global increase of delta and theta power also have been seen in AD and mild cognitive impairment in comparison with healthy controls (Moretti et al., 2004; Jeong, 2004; Osipova et al., 2005; Hsiao et al., 2014; Koelewijn et al., 2017). The shifting of power spectrum from high-frequency oscillations as alpha, beta and gamma to low-frequency oscillations as delta and theta commonly reported in AD patients (Dauwels et al., 2010; Vecchio et al., 2013; Cassani et al., 2018). In a limited number of studies comparing EOAD and LOAD, it was found that EOAD patients showed more EEG abnormalities such as slow or sharp waves than LOAD patients and EOAD patients had lower alpha and higher delta power than LOAD patients globally (de Waal et al., 2011, 2012).

Due to EEG's advantages as a neurophysiological method, diagnostic research protocols could include EEG as a low-cost and high sensitivity diagnostic tool. It is reported that EEG markers are able to discriminate individuals with AD from healthy controls with an accuracy of 80% (Buscema et al., 2007; Blinowska et al., 2017). The ratios and combined values are advantageous because absolute powers differ between individuals (Bennys et al., 2001). Combined alpha-theta global powers (Huang et al., 2000) and alpha/theta ratio (Bennys et al., 2001) were described as valuable research tools in discriminating between AD patients and healthy controls with an accuracy rate of 84%. The alpha/theta ratio is a valuable research tool and its diagnostic accuracy rate is shown to be greater than visual inspection (Gueguen et al., 1991). It has been shown that lower alpha/theta ratio indicates decreased alpha and increased theta in AD patients (Cibils, 2002), which is often seen as pathological. Schmidt and colleagues (Schmidt et al., 2013) demonstrated that alpha/theta ratios from O_1 and C_3 electrodes could discriminate between AD patients and healthy controls with 76.4% sensitivity and 84.6% specificity. Many number of studies have been conducted to discriminate individuals with LOAD from healthy controls, however to the best of our knowledge, none has been conducted for EOAD.

In the current study, with the intention of comparing our results with the earlier studies' findings, we used the power analysis. The aim of the present study is to compare EOAD patients with healthy controls and LOAD patients using resting-state EEG. Our goal is to investigate resting-state delta, theta, alpha (alpha1, alpha2) and beta (beta-1 and beta-2) powers of groups to show any possible differences in AD according to age-onset of the disease. In addition, we aimed to provide a cut-off score of alpha/theta ratio in order to discriminate individuals with EOAD and young healthy controls.

In the present study, we hypothesized that individuals with EOAD would show increased power in slow rhythms (i.e., delta and theta frequencies) and decreased power in fast rhythms (i.e., alpha and beta frequencies) in comparison with healthy controls, and their electrophysiological abnormalities which would be more severe in comparison to individuals with LOAD.

2. Materials and Methods

2.1. Participants

The study is conducted with forty-seven EOAD individuals and 51 LOAD individuals. All patients were diagnosed according to the National Institute of Aging-Alzheimer's Association diagnostic guideline (McKhann et al., 2011). The inclusion criteria for AD

patients included: (1) impairment of two or more cognitive domains; (2) impaired daily living activities with CDR score of 1 or 2. The inclusion criteria for the healthy controls were: no history or presence of any neurological and psychiatric abnormalities, no history of drug and alcohol misuse and/or Mini Mental State Examination (MMSE) score of ≥27. The exclusion criteria for AD patients were history or presence of any other neurological and/or psychiatric disorders including depression, traumatic brain injury, vascular brain lesions, and alcohol or drug misuse. All AD patients were on cholinesterase inhibitor drugs (donepezil; 5–10 mg per day and rivastigmine; 6 mg- 9.5 mg/24 h per day). Eleven individuals with LOAD and seven individuals with EOAD were treated with memantine (20 mg per day) in addition to cholinesterase inhibitors. The smoking rate among participants was 24.6% for young HC, 26.7% for older HC, 23.0% for EOAD and 25.7% for LOAD with no statistically significant differences ($X^2_{3,191}$ = 6.753p = 0.080). All participants underwent a comprehensive neuropsychological test battery, neuroimaging, and neurological examination. AD diagnosis was confirmed in four individuals with LOAD and 21 individuals with EOAD by CSF-AD biomarkers (Park et al., 2019). All dementia cases (n = 98) had Clinical Dementia Rating (CDR) score of 1 except for 2 LOAD and 3 EOAD patients with CDR 2. The independent t-test revealed no differences in the CDR scores between EOAD and LOAD [(t 96) = 0.653p = 0.515]. Forty-nine young and 51 older healthy participants matched to individuals with EOAD and LOAD in terms of age, education and gender were also included in the study. All participants provided written informed consent according to the Helsinki declaration. This study is approved by the local ethics committee of Dokuz Eylül University. Table 1 presents the demographic information of the participants.

2.2. Neuropsychological measures

All participants underwent a comprehensive neuropsychological test battery which included Oktem Verbal Memory Processes Test (OVMPT), WMS-R Digit Span Test, Stroop Test, simple copying tests, Clock Drawing Test, verbal fluency tests (category and phonemic), Boston Naming Test, and Mini Mental State Examination (MMSE). The raw scores from neuropsychological tests were transformed into z-scores using normative mean and standard deviation. Then, the composite scores were elicited to from five cognitive domains; memory (OVMPT total score, delayed recall, free recall and total recall), attention (Digit Span Test-Forward, WORLD spelled backward), visuospatial skills (simple copying tests, Clock Drawing Test), executive functions (Stroop Test, phonemic verbal fluency test, categorical verbal fluency test, Digit Span Test-Backward) and language (Boston Naming Test).

Values were presented as mean ± standard deviation. HC: healthy controls; EOAD: early-onset Alzheimer's Disease; LOAD: late-onset Alzheimer's Disease; MMSE: Mini Mental State Examination. The sign "*" shows the statistical difference between EOAD and LOAD; the sign "†" shows the statistical difference between EOAD and young HC; the sign "‡" shows the statistical difference between EOAD and older HC. [a]One-Way ANOVA; [b]Chi-square analysis.

The ANCOVA on cognitive domains showed the neither effect of age nor education as covariates. Thus, these factors are not used in further analysis.

Results of the ANOVA revealed main GROUP effect in MMSE scores ($F_{3,195}$ = 144.998; $p < 0.001$), memory ($F_{3,193}$ = 381.677; $p < 0.001$), attention ($F_{3,118}$ = 6.913; $p < 0.001$), visuospatial skills ($F_{3,158}$ = 53.721; $p < 0.001$), executive functions ($F_{3,101}$ = 6.976; $p < 0.001$) and language domains ($F_{3,188}$ = 15.121; $p < 0.001$). Further analysis indicated that both AD groups showed significantly lower scores than HC groups in MMSE scores, memory, visuospatial skills, language and attention domains (all, $p < 0.015$). In addition, individuals with EOAD had lower scores in the visuospatial domain in comparison with individuals with LOAD ($p < 0.001$).

2.3. EEG recordings & analysis

EEG was recorded from 30 Ag/AgCl electrodes placed on an elastic cap (Easy-Cap) according to the international 10–20 system and linked ear electrodes (A1 + A2) were used as reference electrodes. Electrooculogram was recorded from the medial upper and lateral orbital location of the right eye. All electrode impedances were kept below 10 kΩ. The EEG was digitized at a rate of 500 Hz with the BrainAmp 32-channel DC amplifier with a 0.03–70 Hz band limit. Resting-state EEG recordings were performed with eyes closed for four minutes during morning hours (approximately between 09.00 to 10.00 am) of typical weekdays.

Offline analyses were performed using Brain Products Vision Analyzer 2.1 (Brain Products GmbH; Gilching, Germany). Continuous EEG recordings were primarily filtered with a 0.01–50 Hz band-pass filter and 12 dB/octave slope, then segmented into 2000 ms epochs. Artifacts were semi-automatically eliminated based on following criteria: (1) maximum amplitude in an epoch was ± 200 μV, (2) maximum allowed voltage step was 50 μV, (3) maximum allowed difference in the segment was 200 μV, and 4) the lowest allowed activity in a 100 ms interval was 0.5 μV. Remaining artifacts were eliminated by excluding inappropriate epochs following visual inspection. The minimum of 85 artifact-free epochs included the analysis from each participant.

A fast Fourier transform (FFT) was applied to all artifact-free epochs, using maximum resolution power, no window function, non-complex output on the half spectrum with a resolution of 0.48 Hz. Artifact-free epochs were zero-padded for a length of 2048 ms generating 1024 points, by adding 48 zeros at the beginning and end of each. The peak power for each frequency band (delta: 0.5–3.8 Hz, theta: 3.9–7.8 Hz, alpha: 8–12.6 Hz, beta: 12.7–30 Hz) was calculated. In addition, alpha and beta frequency bands are divided as fast and slow frequencies. Alpha-1 (8–10.25 Hz), alpha-2 (10.26–12.6 Hz), beta-1 (12.7–20 Hz) and beta-2 (20.1–30 Hz) peak power measurements were performed. The FFT was applied on each artifact-free epoch and then averaged. In addition, the peak power measurements were performed without any artifact rejection, as it can be seen in Supplementary

Table 1
Demographics and clinical data of participants.

	Age < 65		Age ≥ 65	
	EOAD	HC	LOAD	HC
N	47	49	51	51
Age[a*]	57.53 ± 3.99	57.53 ± 4.07	74.90 ± 4.90	73.94 ± 5.61
Education[a†‡]	9.04 ± 4.42	12.55 ± 4.38	8.55 ± 4.40	11.13 ± 4.81
Gender (F/M)[b]	29/18	28/21	28/23	25/26
MMSE[a†‡]	18.62 ± 5.07	29.14 ± 0.98	20.71 ± 4.05	28.96 ± 1.17
pTau181/Aß(1–42) pg/mL	0.152 ± 0.06	-	-	-

Methods. The alpha-1, alpha-2, beta-1 and beta-2 sub-bands have been chosen for better comparison with the previous dementia literature (Babiloni et al., 2006a) and for better understanding of 2 components of alpha rhythm: (1) high alpha (10–13 Hz) which considered to be reflect neural activity associated with the sense and elaboration of meaning-related information; (2) the lower alpha (8–10 Hz) is reported involved in mental anticipation (Ishii et al., 2017; Klimesch, 1999; Babiloni et al., 2006b).

Since we do not expect any hemispheric differences, the 12 electrodes were grouped based on anterior-posterior electrode locations, as frontal (F_3, F_z, F_4), central (C_3, C_z, C_4), temporal (T_7, T_8, TP_7, TP_8, P_7, P_8), parietal (P_3, P_z, P_4), and occipital (O_1, O_z, O_2). Spectral power values for each frequency band were averaged across aforementioned channels.

2.4. Statistical analysis

IBM SPSS Statistics 24.0 software (SPSS Inc., Chicago, USA) was used for all statistical analyses. One-way ANOVA was used for the analysis of demographic information and neuropsychological data. Separate repeated measures ANCOVAs were conducted for each frequency band. Repeated measures ANCOVAs with within-subject factor 5-level Location (Frontal, Central, Temporal, Parietal, and Occipital) and between-subject factor 4-level Group (EOAD, LOAD, Young HC, Older HC) was performed on the peak power, with age and education as covariates. In order to avoid unnecessary comparisons between groups, the factor group was analyzed by using contrasts. EOAD vs. LOAD, Young HC vs. EOAD, Older HC vs. LOAD and Young vs. Older HC were compared as planned *a priori*. The Bonferroni correction was applied to all non-planned post-hoc comparisons and corrected *p* values were reported. Receiver operating characteristic (ROC) curve analysis was conducted to evaluate diagnostic accuracy of alpha/theta ratio from frontal, central, parietal and occipital locations. In addition, the frequency bands which showed a significant difference among groups were analyzed by ROC curve analysis. In order to determine cut off scores, Youden Indices were calculated. The level of $p < 0.05$ was considered significant for all comparisons. We also conducted a post hoc power analysis with the SPSS for calculating power of ANOVAs and G*Power (Erdfelder et al., 1996). See Supplementary Results for details (Figure Supplementary figure 1). In addition, supplementary repeated measures ANOVA and ROC analysis on peak power in delta, theta, alpha, beta, alpha-1, alpha-2, beta-1, beta-2 frequency bands were conducted. See Supplementary Results for details.

3. Results

The repeated measures ANCOVA on peak power in delta, theta, alpha, beta, alpha-1, alpha-2, beta-1, beta-2 frequency bands showed neither the impact of age nor the education. Thus, these factors were not used as covariates in further analysis. The unmentioned effects were not statistically significant. The Figure 1 presents frequency band differences at all locations across groups.

3.1. The peak power in the delta frequency

The main GROUP effect ($F_{3,194} = 11.611$; $p < 0.001$) showed that individuals with EOAD had higher delta power than individuals with LOAD and young HC (all, $p < 0.001$) (Figure 2A).

There was a LOCATION × GROUP interaction effect ($F_{3,582} = 3.323$; $p = 0.008$) indicating that individuals with EOAD showed higher theta power than young HC and in frontal ($p = 0.007$, $p < 0.001$), central ($p = 0.003$, $p = 0.001$), parietal (all, $p < 0.001$), and occipital (all, $p < 0.001$) electrode locations.

3.2. The peak power in the theta frequency

The main GROUP effect ($F_{3,194} = 22.878$; $p < 0.001$) showed that individuals with EOAD had higher theta power than individuals with LOAD and young HC (all, $p < 0.001$). In addition, post-hoc analysis revealed that individuals with LOAD showed higher theta power than older HC ($p < 0.001$) (Figure 2B).

There was a LOCATION × GROUP interaction effect ($F_{3,582} = 8.294$; $p < 0.001$) indicating that individuals with EOAD showed higher theta power than LOAD in frontal ($p < 0.001$), central ($p < 0.001$) and parietal ($p = 0.001$) electrode locations and young HC in all locations (all, $p < 0.001$).

3.3. The peak power in the alpha frequency

The main GROUP effect was found in the alpha frequency band ($F_{3,194} = 6.434$; $p < 0.001$). Post-hoc analysis showed that individuals with EOAD showed reduced alpha power than young HC ($p < 0.001$) (Figure 2C).

3.4. The peak power in the alpha-1 frequency

The main GROUP effect was found in the alpha-1 frequency band ($F_{3,194} = 6.139$; $p = 0.001$). Post-hoc analysis showed that individuals with EOAD showed reduced alpha-1 power than young HC ($p < 0.001$) (Figure 2C).

3.5. The peak power in the alpha-2 frequency

There were main GROUP and LOCATION × GROUP interaction effects in the alpha-2 frequency ($F_{3,194} = 8.592$; $p < 0.001$; $F_{12,776} = 2.605$; $p = 0.015$). Post-hoc analysis revealed that individuals with EOAD showed reduced alpha-2 power than young HC in frontal, central, temporal, and parietal locations (all, $p < 0.010$). Furthermore, young HC had higher alpha-2 power than older HC in all electrode locations (all, $p < 0.007$) Figure 2(C).

3.6. The peak power in the beta frequency

There was a main GROUP effect in the beta frequency ($F_{3,194} = 4.459$; $p = 0.005$) in which young HC had higher beta power than older HC ($p = 0.004$) and individuals with EOAD ($p = 0.001$) (Figure 2D).

3.7. The peak power in the beta-1 frequency

There was a main GROUP effect in the beta-1 frequency ($F_{3,194} = 4.576$; $p = 0.004$) in which young HC had higher beta-1 power than older HC ($p = 0.002$) and individuals with EOAD ($p = 0.001$) (Figure 2D).

3.8. The peak power in the beta-2 frequency

There was a main GROUP effect in the beta-2 frequency ($F_{3,194} = 4.820$; $p = 0.003$), indicating young HC had higher beta-2 power than older HC ($p = 0.034$) and individuals with EOAD ($p < 0.001$) (Figure 2D).

3.9. ROC analyses

3.9.1. EOAD vs. Young HC

The ROC curves showed that frontal (AUC = 0.881, $p < 0.001$), central (AUC = 0.882, $p < 0.001$), temporal (AUC = 0.881, $p < 0.001$), parietal (AUC = 0.877, $p < 0.001$), and occipital (AUC = 0.846, $p < 0.001$) alpha/theta ratios can discriminate individuals with EOAD and young healthy controls (Figure 3A, Table 2).

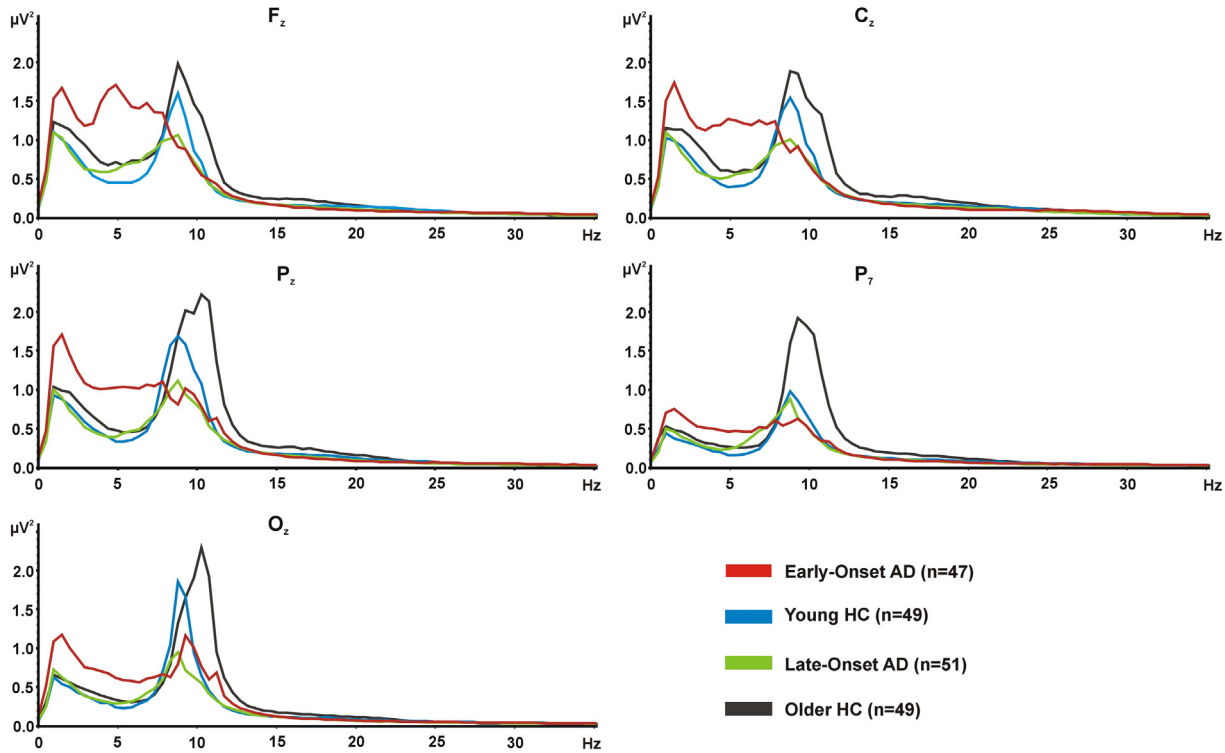

Fig. 1. Grand averages of delta, theta, alpha and beta powers across groups at frontal, central, parietal, temporal, and occipital electrode locations. The F_Z, C_Z, P_Z, P_7, and O_Z electrodes were selected respectively for frontal, central, parietal, temporal, and occipital locations.

The ROC analysis also showed that frontal (AUC = 0.882, p < 0.001), central (AUC = 0.884, p < 0.001), temporal (AUC = 0.879, p < 0.001), parietal (AUC = 0.885, p < 0.001), and occipital (AUC = 0.847, p < 0.001) alpha1/theta ratios; and frontal (AUC = 0.882, p < 0.001), central (AUC = 0.867, p < 0.001), temporal (AUC = 0.880, p < 0.001), parietal (AUC = 0.851, p < 0.001), and occipital (AUC = 0.823, p < 0.001) alpha2/theta ratios can discriminate EOAD and young HC (Table 2).

The results of ROC analyses showed that central (AUC = 0.650, Sen = 46.8%, Spe = 87.5%, cut off = 1.46, YI = 0.33, p = 0.011), parietal (AUC = 0.650, Sen = 70.2%, Spe = 38.85%, cut off = 0.90, YI = 0.09, p = 0.003), and occipital delta (AUC = 0.724, Sen = 57.4%, Spe = 83.7%, cut off = 1.00, YI = 0.41, p < 0.001); and frontal (AUC = 0.840, Sen = 68.1%, Spe = 89.8%, cut off = 1.13, YI = 0.58, p < 0.001), central (AUC = 0.807, Sen = 87.2%, Spe = 65.3%, cut off = 0.65, YI = 0.53, p < 0.001), temporal (AUC = 0.781, Sen = 68.1%, Spe = 83.7%, cut off = 0.90, YI = 0.09, p < 0.001), parietal (AUC = 0.800, Sen = 70.2%, Spe = 81.6%, cut off = 0.69, YI = 0.52, p < 0.001), and occipital theta (AUC = 0.774, Sen = 83.0%, Spe = 69.4%, cut off = 0.46, YI = 0.52, p < 0.001) power can discriminate EOAD and young HC.

The ROC analysis also indicated that frontal (AUC = 0.696, Sen = 80.9%, Spe = 63.3%, cut off = 1.19, YI = 0.44, p = 0.001), central (AUC = 0.705, Sen = 63.8%, Spe = 75.5%, cut off = 0.83, YI = 0.39, p = 0.001), temporal (AUC = 0.784, Sen = 78.7%, Spe = 65.3%, cut off = 0.68, YI = 0.44, p < 0.001), parietal (AUC = 0.733, Sen = 89.4%, Spe = 51.0%, cut off = 2.20 YI = 0.40, p < 0.001), and occipital alpha (AUC = 0.742, Sen = 78.7%, Spe = 63.3%, cut off = 1.19, YI = 0.42, p < 0.001); and frontal (AUC = 0.640, Sen = 69.6%, Spe = 57.1%, cut off = 0.24, YI = 0.27, p = 0.019), central (AUC = 0.661, Sen = 41.3%, Spe = 77.6%, cut off = 1.15, YI = 0.19, p = 0.007), temporal (AUC = 0.687, Sen = 93.5%, Spe = 42.9%, cut off = 0.25, YI = 0.36, p = 0.002), parietal (AUC = 0.699, Sen = 54.3%, Spe = 73.5%, cut off = 1.17 YI = 0.28, p = 0.001), and occipital beta (AUC = 0.621, Sen = 93.5%, Spe = 32.7%, cut off = 0.40, YI = 0.26, p = 0.042) power can discriminate EOAD and young HC.

3.9.2. LOAD vs. Old HC

The ROC curves showed that frontal (AUC = 0.688, p = 0.001), central (AUC = 0.698, p = 0.001), temporal (AUC = 0.659, p = 0.006), parietal (AUC = 0.708, p < 0.001), and occipital (AUC = 0.679, p = 0.001) alpha/theta ratios can discriminate individuals with LOAD and old healthy controls (Figure 3B, Table 3). The ROC analysis also revealed that frontal (AUC = 0.680, p = 0.002), central (AUC = 0.684, p = 0.001), temporal (AUC = 0.659, p = 0.006), parietal (AUC = 0.689, p = 0.001), and occipital (AUC = 0.659, p = 0.009) alpha1/theta ratios; and frontal (AUC = 0.645, p = 0.012), central (AUC = 0.622, p = 0.034), temporal (AUC = 0.664, p = 0.004), parietal (AUC = 0.647, p = 0.010), and occipital (AUC = 0.636, p = 0.018) alpha2/theta ratios can discriminate LOAD and old HC (Table 3).

The ROC analysis also indicated that frontal (AUC = 0.667, Sen = 68.6%, Spe = 64.7%, cut off = 1.50, YI = 0.33, p = 0.004), central (AUC = 0.633, Sen = 68.6%, Spe = 64.7%, cut off = 1.74, YI = 0.31, p = 0.021), temporal (AUC = 0.658, Sen = 66.7%, Spe = 60.8%, cut off = 2.44, YI = 0.28, p = 0.006), parietal (AUC = 0.619, Sen = 74.5%, Spe = 64.7%, cut off = 2.36 YI = 0.39, p = 0.038) and occipital theta (AUC = 0.617, Sen = 68.6%, Spe = 60.8%, cut off = 2.16, YI = 0.29, p = 0.042) can discriminate LOAD and old HC.

3.10. EOAD vs. LOAD

The ROC curves showed that frontal (AUC = 0.768, p < 0.001), central (AUC = 0.767, p < 0.001), temporal (AUC = 0.767, p < 0.001), parietal (AUC = 0.759, p < 0.001), and occipital (AUC = 0.676, p = 0.003) alpha/theta ratios can discriminate individuals with EOAD and LOAD (Figure 3C, Table 4). The ROC analysis

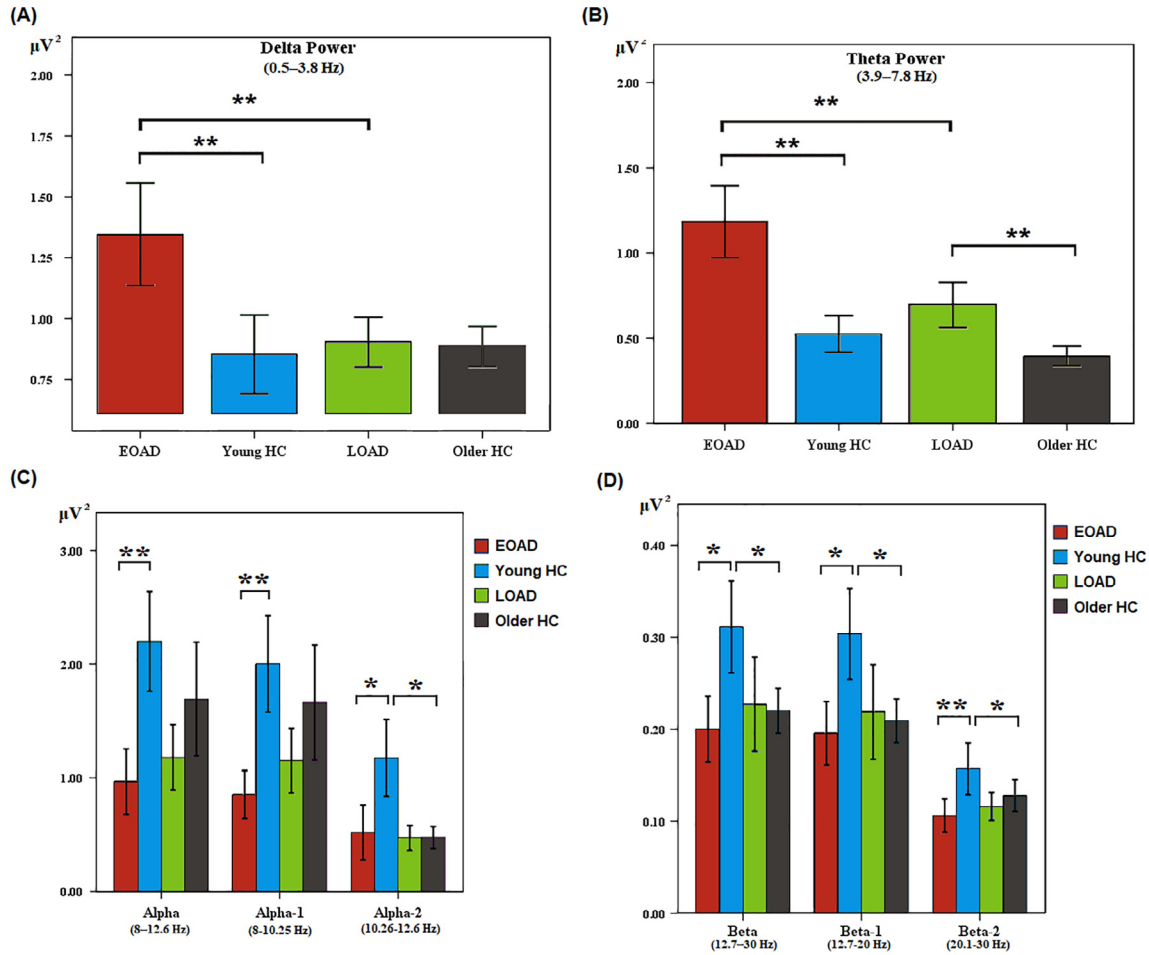

Fig. 2. (A) Delta powers across groups; (B) theta powers across groups; (C) alpha, alpha-1, and alpha-2 powers across groups; (D) beta, beta-1, and beta-2 powers across groups. EOAD: Early-onset Alzheimer's disease; LOAD: Late-onset Alzheimer's disease; HC: Healthy control. * $p < 0.05$, ** $p < 0.001$.

also showed that frontal (AUC = 0.779, $p < 0.001$), central (AUC = 0.780, $p < 0.001$), temporal (AUC = 0.768, $p < 0.001$), parietal (AUC = 0.788, $p < 0.001$), and occipital (AUC = 0.755, $p < 0.001$) alpha1/theta ratios and frontal (AUC = 0.769, $p < 0.001$); central (AUC = 0.746, $p < 0.001$), temporal (AUC = 0.777, $p < 0.001$), parietal (AUC = 0.753, $p < 0.001$), and occipital (AUC = 0.740, $p < 0.001$) alpha2/theta ratios can discriminate EOAD and LOAD (Table 4).

The ROC analysis also indicated that frontal (AUC = 0.683, Sen = 93.6%, Spe = 39.2%, cut off = 0.95, YI = 0.33, p = 0.002), central (AUC = 0.685, Sen = 44.7%, Spe = 84.3%, cut off = 1.52, YI = 0.29, p = 0.002), parietal (AUC = 0.716, Sen = 70.2%, Spe = 68.6%, cut off = 1.04 YI = 0.39, $p < 0.001$), and occipital delta (AUC = 0.686, Sen = 63.8%, Spe = 72.5%, cut off = 0.93, YI = 0.36, p = 0.002); and frontal (AUC = 0.758, Sen = 78.7%, Spe = 62.7%, cut off = 1.00, YI = 0.42, $p < 0.001$), central (AUC = 0.725, Sen = 87.2%, Spe = 52.9%, cut off = 0.64, YI = 0.40, $p < 0.001$), temporal (AUC = 0.676, Sen = 68.6%, Spe = 64.7%, cut off = 1.50, YI = 0.33, p = 0.003), parietal (AUC = 0.704, Sen = 89.4%, Spe = 45.1%, cut off = 0.53 YI = 0.35, $p < 0.001$), and occipital theta (AUC = 0.660, Sen = 93.6%, Spe = 37.3%, cut off = 0.33, YI = 0.31, p = 0.006) can discriminate EOAD from LOAD.

4. Discussion

The aim of the current study was to investigate the differentiation of resting-state EEG dynamics in individuals with AD according to the age of onset in comparison to demographically-matched healthy controls. In our study, we found that individuals with EOAD showed increased delta and theta power with decreased alpha and beta resting-state EEG power compared to young healthy controls. They also displayed increased delta and theta power in comparison with LOAD. Additionally, we once more showed that theta power increased in LOAD compared to older healthy controls, and fast frequencies (alpha and beta) decreased in healthy aging.

4.1. Peak power results

Our hypotheses were confirmed by our results, which showed that EOAD patients display more widespread and severe electrophysiological abnormality than individuals with LOAD and demographically-matched healthy controls. The EOAD group shows increased delta and theta power, and decreased alpha and beta power than aged-matched healthy controls in frontal, central and parietal locations. Previously, De Waal and his colleagues (De Waal et al., 2012) showed that increased delta and theta power in EOAD compared to LOAD patients. When comparing EOAD with healthy controls, the most common finding in the literature is increased slow-wave activity in EOAD compared to controls (Micanovic and Pal, 2014). In line with this, an increment in delta and theta global field power, in addition to a decrease in alpha and beta global field power was previously found in AD (Huang et al., 2000). Global delta and beta field power showed an inverse association with p-tau and total tau (Smailovic et al., 2018). Also, recently Gaubert et al. (2019) reported that effects of neurodegeneration can be seen as the most conspicuously in fronto-central

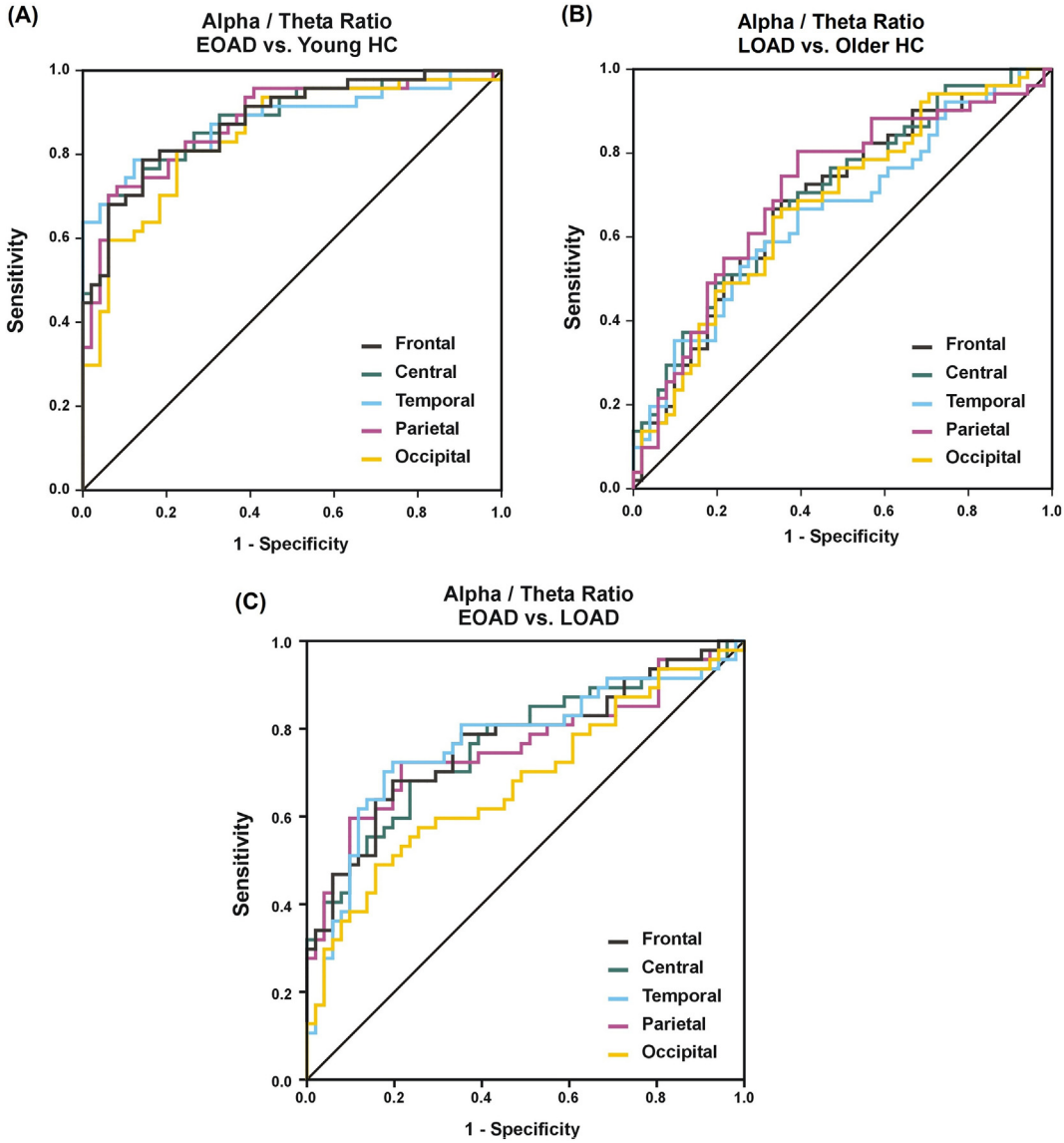

Fig. 3. (A) ROC curves of alpha/theta ratio across early-onset Alzheimer's disease and young healthy controls at frontal, central, parietal, temporal, and occipital locations; (B) ROC curves of alpha/theta ratio across late-onset Alzheimer's disease and older healthy controls at frontal, central, parietal, temporal, and occipital locations; (C) ROC curves of alpha/theta ratio across early-onset and late-onset Alzheimer's disease at frontal, central, parietal, temporal, and occipital locations. EOAD: Early-onset Alzheimer's disease, LOAD: Late-onset Alzheimer's disease, HC: Healthy control.

areas with increased high frequencies and decreased low frequencies.

Focal generators of slow waves, which are often seen as global processing modes, had significant importance in pathological aging (Ishii et al., 2017). Alterations in slow frequencies as a result of cholinergic dysfunction might be mediated by amyloid pathology (Smailovic et al., 2018; Smailovic and Jelic, 2019). Previous studies demonstrated slow-wave activity is a predictor of cognitive status and linked to the rate of atrophy in AD patients (Fernandez et al., 2003, 2002).

Resting-state delta rhythms are associated with loss of gray matter in AD (Babiloni et al., 2015, 2013). Also, it is shown that long-term use of Acetylcholinesterase inhibitors normalizes the alpha and delta rhythms to a certain extent (Rodriguez et al., 2002; Babiloni et al., 2006b, 2020). It is claimed that the delta frequency may reflect the regulation of vigilance which involves subcorticocortical, corticothalamic, and corticocortical neural circuits (Pfurtscheller and Lopez da Silva, 1999; Wan et al., 2019; Babiloni et al., 2020), and the disruption in delta, may show the underlying pathophysiological processes reflecting impaired functional connectivity among thalamocortical and corticothalamic circuits (Klimesch, 1999; Babiloni et al., 2020). In AD patients, the amplitude of delta EEG sources correlated with the frontal white matter volume, which may result by loss and/or dysregulation of cholinergic basal forebrain neurons projecting to the hippocampus and fronto-parietal connections (Jeong, 2004; Babiloni et al., 2006a). Increment in delta activity might indicate higher cholinergic loss (Pucci et al., 1999), which was previously found associated with EOAD (Riekkinen et al., 1991). Thus, the increase in delta power can be speculated as individuals with EOAD had more cholinergic deficits than individuals with LOAD (Eckerström et al., 2018). In addition, several studies have shown that individuals with EOAD had larger structural damage with greater metabolic and perfusion deficits in neocortex, whereas individuals with LOAD shows gray matter loss and metabolic changes in medial temporal area (Frisoni et al., 2005, 2007; Ossenkoppele et al., 2012; Gour et al., 2014). Kalpouzos et al. (2005) found frontal and parietal hypometabolism and Adriaanse et al. (2014) reported

Table 2
Sensitivity, specificity, cut off scores and Youden indices of ROC Curve Analyses in frontal, central, temporal, parietal and occipital electrode locations for early-onset Alzheimer's disease vs. young healthy controls.

	Alpha/Theta Ratio				
	Frontal	Central	Temporal	Parietal	Occipital
Sensitivity (%)	81	71	79	83	81
Specificity (%)	82	94	88	75	78
Cut off	1.07	0.72	1.15	1.75	2.02
Youden Index	0.62	0.64	0.67	0.59	0.58
	Alpha1/Theta Ratio				
	Frontal	Central	Temporal	Parietal	Occipital
Sensitivity (%)	79	70	66	77	68
Specificity (%)	86	94	96	90	88
Cut off	0.89	0.72	0.78	0.87	1.08
Youden Index	0.64	0.64	0.62	0.66	0.56
	Alpha2/Theta Ratio				
	Frontal	Central	Temporal	Parietal	Occipital
Sensitivity (%)	89	83	79	68	68
Specificity (%)	73	82	84	98	88
Cut off	0.58	0.58	0.71	0.39	0.55
Youden Index	0.63	0.65	0.63	0.66	0.59

ROC: receiver operating characteristic.

Table 3
Sensitivity, specificity, cut off scores and Youden indices of ROC Curve Analyses in frontal, central, temporal, parietal and occipital electrode locations for late-onset Alzheimer's disease vs. older healthy controls.

	Alpha/Theta Ratio				
	Frontal	Central	Temporal	Parietal	Occipital
Sensitivity (%)	69	71	67	80	67
Specificity (%)	65	61	61	61	65
Cut off	1.50	1.78	2.45	2.65	1.98
Youden Index	0.33	0.31	0.28	0.41	0.31
	Alpha1/Theta Ratio				
	Frontal	Central	Temporal	Parietal	Occipital
Sensitivity (%)	71	51	67	82	65
Specificity (%)	61	76	61	55	63
Cut off	1.52	1.30	2.45	2.71	1.86
Youden Index	0.31	0.28	0.28	0.37	0.27
	Alpha2/Theta Ratio				
	Frontal	Central	Temporal	Parietal	Occipital
Sensitivity (%)	84	57	65	71	78
Specificity (%)	41	63	61	57	49
Cut off	1.09	0.82	1.07	1.07	1.28
Youden Index	0.26	0.20	0.26	0.28	0.28

ROC: receiver operating characteristic.

more widely impaired functional organization in EOAD compared to LOAD. In this line, our findings may be the EEG reflection of greater functional impairment in EOAD. Besides, Başar and his colleagues (Başar et al., 2016) showed that event-related delta amplitudes are low in healthy individuals during quiet wakefulness, however they are enhanced rapidly during a sensory event which triggers global arousal. Decreased delta power in EOAD compared to LOAD and healthy controls may also reflect more densely impaired connectivity in neural circuits, vigilance, and arousal.

Previous studies revealed an increase in theta power in individuals with AD compared to healthy controls (Babiloni et al., 2020). It was shown that the AD group with age of onset below 60 years, had a propensity to a greater 1–6.5 Hz EEG activity compared to older individuals with AD (Pucci et al., 1999). Theta power was found increased in EOAD compared to LOAD (Schreiter-Gasser et al., 1994) and general theta increment was related to global cognitive decline and slowing of cognitive speed (Claus et al., 1998), which may be an early marker of neurodegeneration in the brain (Stomrud et al., 2010). Additionally, it is shown that some CSF markers for AD are associated with relative theta power suggesting an increase in theta power that reflects a decrease in cognitive speed (Stomrud et al., 2010). Musaeus and colleagues (Musaeus et al., 2018) showed that relative theta power had the highest correlation with total tau and suggested that theta power in eyes-closed condition would be a hallmark in early indication of neurodegeneration. Theta power is also found correlated with hippocampal atrophy rate (Grunwald et al., 2007; Moretti et al., 2009). According to Bastiaansen and Hagoort (2003), the cortical theta activity reflects the associations between cortical and hippocampal generators of theta. Thus, EEG theta power is thought to reflect degenerative changes in the cortico-hippocampal connections, and shows the underlying network dysfunction (Klimesch, 1997). Although it is known that individuals with LOAD had more atrophy and disrupted connectivity in the hippocampus (Park et al., 2017), we may speculate that age of onset may affect the cortical connectivity differently in AD. It may be thought that hippocampal-cortical connections in EOAD are more impaired than LOAD patients due to more widespread, severe atrophy, and

Table 4
Sensitivity, specificity, cut off scores and Youden indices of ROC Curve Analyses in frontal, central, temporal, parietal, and occipital locations for early-onset vs. late-onset Alzheimer's disease.

	Alpha/Theta Ratio				
	Frontal	Central	Temporal	Parietal	Occipital
Sensitivity (%)	68	68	72	72	57
Specificity (%)	80	76	80	78	74
Cut off	0.57	0.67	0.94	0.78	0.86
Youden Index	0.49	0.45	0.53	0.51	0.32
	Alpha1/Theta Ratio				
	Frontal	Central	Temporal	Parietal	Occipital
Sensitivity (%)	59	80	65	76	73
Specificity (%)	80	62	77	68	66
Cut off	0.91	1.66	1.68	1.76	1.92
Youden Index	0.42	0.42	0.42	0.45	0.40
	Alpha2/Theta Ratio				
	Frontal	Central	Temporal	Parietal	Occipital
Sensitivity (%)	68	64	52	73	51
Specificity (%)	74	77	89	65	87
Cut off	0.50	0.62	0.58	1.03	0.50
Youden Index	0.42	0.41	0.41	0.39	0.38

ROC: receiver operating characteristic.

impaired connectivity in the neocortex, and/or CSF pathology in EOAD.

Fast frequency bands which presumably generated in the cortex might be especially affected by degeneration and neuronal loss associated with tau pathology (Smailovic et al., 2018; Smailovic and Jelic, 2019). Several studies showed that alpha oscillations are generated by thalamo-cortical networks and modulated by brainstem cholinergic neurons (Osipova et al., 2005). Alpha band is associated with many basic cognitive functions, which regulate higher cognitive functions. Thus, the impairment of alpha oscillations may indicate that from basic to complex EOAD patients had major cognitive impairments. It is known that individuals with EOAD had more pathological burden than healthy individuals and LOAD (Marshall et al., 2007). In line with the literature suggesting cholinergic deficits underlie the cognitive and pathological changes in AD, abnormalities in alpha power may reflect the pathology of the cholinergic system. The results of our study may reflect more pronounced pathological burden and cholinergic deficits in EOAD. Further research is needed to investigate the correlation of alpha power and pathology in the cholinergic system. Even though findings of other frequency bands vary, alpha band findings are highly consistent in the literature showing that individuals with AD and MCI had lower alpha power compared to healthy controls (Ishii et al., 2017; Babiloni et al., 2020). Similar to delta EEG activity, alpha rhythms are also associated with atrophy of gray matter and AD-related genetic factors. In addition, alpha rhythms are normalized by administering Acetylcholinesterase inhibitors (Babiloni et al., 2020). This may be the reason for insignificant alpha results in the current study between EOAD and LOAD, since all participants with AD were receiving cholinesterase inhibitor drugs. Moreover, several studies reported that slow frequency bands (delta and theta) disrupted before fast frequency bands (alpha and beta), and some claimed that the theta activity is the best discriminating electrophysiological parameter between AD, mild cognitive impairment and healthy individuals (Jelic et al., 1996, Huang et al., 2000; Grunwald et al., 2001; Ishii et al., 2017). Our study shows no difference in alpha and beta band between EOAD and LOAD, most likely due to the fact that changes in slow bands, especially in theta, are more pronounced than both beta and alpha in the early phase of the disease.

Even though beta oscillations are considered to be associated with movement-related activities rather than cognitive functions (Honaga et al., 2010), recent studies showed that beta power had an important role in working memory processing (Pesonen et al., 2006, 2007). Besides, several studies showed that beta rhythms are positively correlated with cognitive functions (Claus et al., 1998; Babiloni et al., 2004; Kaiser et al. 2017), and alertness (Gola et al., 2012). A study specifically found that beta rhythm is strongly correlated with memory function (Kaiser et al., 2017). While there is no sufficient evidence about the role of beta in cognition, our findings may suggest that the decreased beta power in EOAD may result from fronto-parietal connections which mediate working memory processes (Kurimoto et al., 2012). Hence, it may be speculated that decreased beta in EOAD, which has not yet been reflected in LOAD in the current study, may be due to the severity of the working memory dysfunction.

Our results showing differentiation across healthy groups, indicate that aging only affects higher frequency bands (alpha-2, beta and both sub-beta frequency ranges), since the alpha is the most affected frequency by aging (Ishii et al., 2017). Our alpha findings were in consonance with earlier studies. Babiloni and colleagues (Babiloni et al., 2006a) have found alpha magnitude decreased by physiological aging and correlates with global cognition. Alpha band reflects mainly inhibition and perception. It was reported that higher alpha is associated with mostly sense and elaboration of meaning related information, and lower alpha is related to mental anticipation (Klimesch, 1997, 1999). Alpha oscillations are thought to be driven by thalamo-cortical circuits (Pfurtscheller, 1992). Earlier cross-sectional studies indicated that thalamic volumes of older healthy individuals were smaller in comparison with younger participants (Fama and Sullivan, 2015). Hughes et al. (2012) reported age-related distributed thalamic connections, especially for thalamo-frontal and thalamo-parietal networks. Inhibition processes also deteriorate over time with aging. Additionally, Vlahou et al. (2014) showed that a decreased alpha activation during physiological aging was associated with cholinergic basal forebrain system that affects thalamus-cortical and cortico-cortical processing. In this context, we can speculate that the difference between alpha power in young and older HC may be the result of age-related differences in thalamic connections. Moreover, according to Klimesch (2007, 2011, 2012) resting-state alpha

band activity is associated with inhibitory filters. The higher the alpha power, the greater a person's ability to build up an effective filter (MacLean and Arnell, 2011). Our results may suggest that in addition to whole inhibition processes, also this filter becomes weaker in aging. Further studies should examine the correlation between thalamic connections and alpha power.

Earlier studies generally suggested that physiological aging has been related to slowing of alpha activity, and enhanced delta, theta and beta activities (Babiloni et al., 2006a; Vlahou et al., 2014). However, we did not find a slow wave difference between healthy groups. There are some longitudinal studies that reported only an increment in delta activity and decrement in beta activity over a 2.5-year follow-up (Coben et al., 1985), while others demonstrated increased beta and theta activities with decreased slow alpha activity with aging, over a 9-year period (Nakano et al., 1992). Although the literature indicates that a slowing of EEG activity and decrement in alpha oscillation with aging, there is no consensus on how aging affects EEG activities, especially for the slow waves (Ishii et al., 2017; Mizukami and Katada, 2018). De Waal and colleagues (De Waal et al., 2012) showed that younger healthy controls had higher beta power in parietal and temporal areas, but lower power in the frontal area than older controls. They also indicated that young healthy controls had higher theta power in frontal, temporal, and parietal areas. This discrepancy between our and aforementioned study may be due to the healthy control group in the aforementioned study which consists of participants who had subjective memory complaints, which is considered as a predictor of developing dementia (Schmand et al., 1996; Kurt et al., 2011; Mitchell et al., 2014).

The current study revealed that lower beta power in older healthy participants in comparison to young healthy individuals. Previous studies reported indifferent or increased beta activity in healthy aging (Babiloni et al., 2006a; Rossiter et al., 2014; Vlahou et al., 2014; Barry and De Blasio, 2017; Moezzi et al., 2019; Kanokwan et al., 2019; Xifra-Porxas et al., 2019). This discrepancy may be due to an age range of participants that were included in these studies. In earlier studies, younger healthy participants' average age was between 19–50 years, while older participants' ages were between 50–85 years. Dustman et al. (1993) pointed out in their review that beta activity increases up to the age of 60, and then starts to decrease. In a recent study investigating age effects, Zhong and Chen (2020) reported lower beta power but, higher beta fMRI-EEG power ratio in the young group than the older group. The authors stated that lowered neurovascular coupling ratio with age might be the cause of higher fMRI-EEG beta power ratio (Zhong and Chen, 2020). In this line, Cook et al. (1998) reported that beta activity was in association with metabolic activity in the cortical area, in their EEG and PET study. It can be speculated that decreased beta power in older healthy participants may be related to a decrease of metabolic activity during aging.

4.2. ROC analysis results

In the present study, we found that the sensitivity and specificity values were above 80% for discriminating EOAD and young HC, 60% for LOAD and older HC, and 70% for EOAD and LOAD by using alpha/theta ratios. The sensitivity and specificity values were slightly lower when separating LOAD and older HC, and EOAD and LOAD but, they remained significant. Our study showed that resting-state EEG markers can successfully discriminate not only early and late-onset AD from healthy controls but also from each other.

It is crucial to know that the atypical presentation of EOAD results in 1.6 years' average delay in diagnosis compared to the LOAD. Diagnosing LOAD is relatively easier compared to EOAD, as a result of its typical presentation i.e. memory impairments. However, some types of early-onset dementia, especially EOAD, may be misdiagnosed as psychiatric disorders (van Vliet et al., 2013; Mendez, 2017). To avoid these complications, the accurate diagnosis of EOAD is a matter of grave importance. In our study, it was found that alpha/theta power ratios in all areas can discriminate individuals with EOAD from young HC accurately, as frontal alpha/theta ratio had a sensitivity of 81% and specificity of 82%, indicating resting-state EEG may be clinically useful for diagnostic purposes.

It is known that the alpha/theta ratio is a promising tool and a valuable marker when discriminating individuals with AD from healthy controls (Schmidt et al., 2013). The alpha/theta ratio is known to be the best discriminating parameter between AD and HC, as well as AD and mild cognitive impairment (Huang et al., 2000).

Even though our results indicate that the peak power can separate groups in almost all frequency bands, it showed that the alpha/theta ratio had the highest sensitivity and specificity. Thus, the use of alpha/theta index in diagnosis rather than a single band power alone may provide better results.

While the current AD markers are sometimes painful procedures and have limited availability, they are generally expensive, invasive, and also time-consuming. The implementation of resting-state EEG in diagnostic processes is needed due to EEG's advantages as being cost-effective, widely available and having a high temporal resolution.

The diagnostic accuracy of resting-state power of alpha/theta ratio and other frequency bands should be evaluated with larger samples and with other methods such as coherence and signal complexity analysis.

The current study has some limitations. The early/late onset-AD groups had a limited number of individuals with CSF biomarkers. In majority of our patients, AD diagnosis was based on neurological assessment, neuropsychological tests, and brain MRI scans. Therefore, future studies should include only CSF-proven AD participants, and healthy individuals with CSF markers, in consideration of the fact that accumulation of senile plaques and neurofibrillary tangles starts 20–30 years earlier from disease onset (Ferreira et al., 2014). In addition, later studies could examine both connectivity-based neuroimaging techniques and EEG to a better understanding of underlying mechanisms of altered EEG dynamics in early/late-onset AD and healthy aging.

5. Conclusions

To the best of our knowledge, this is the first study that examines resting-state EEG power in individuals with EOAD compared with a demographically-matched healthy control group. To conclude, the current study indicates the different patterns of resting-state EEG dynamics of early and late-onset AD, and healthy aging. Additionally, our study showed that resting-state alpha/-theta power ratio can accurately discriminate between individuals with EOAD and healthy controls. EOAD is often misdiagnosed as a psychiatric disorder, mostly depression, and resting-state EEG can be useful in differential diagnosis of AD. Further studies should focus on the discrimination of EOAD and geriatric depression. Our findings can be summarized as altered resting-state EEG power of individuals with EOAD compared to young healthy controls, in all frequency bands, and of individuals with LOAD in comparison with older healthy controls in the theta band. High frequencies were found to be influenced by physiological aging. It is important to evaluate the neurophysiological changes of AD with different age of onset to improve the accurate diagnostic techniques, as well as showing disassociation of healthy aging from pathological aging. Clinical implications of resting-state oscilla-

tions have an importance in ensuring the diagnosis of suspected AD, especially in those of early age-onset.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.clinph.2021.05.012.

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